

Constants:

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$k = 9.0 \times 10^9 \text{ N m}^2/\text{C}^2 \quad k = \frac{1}{4\pi\epsilon_0}$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N m}^2)$$

$$\mu_0 = 1.26 \times 10^{-6} \text{ T m/A}$$

$$c = 3.00 \times 10^8 \text{ m/s}$$

$$c^2 = 931.49 \text{ MeV/u} \quad [E] = \text{MeV}$$

$$h = 6.62 \times 10^{-34} \text{ J s}$$

$$= 4.14 \times 10^{-15} \text{ eV s}$$

$$\hbar = 1.05 \times 10^{-34} \text{ J s} = \frac{h}{2\pi} \quad E = \hbar\omega$$

$$= 6.58 \times 10^{-16} \text{ eV s}$$

$$m_e = 9.1 \times 10^{-31} \text{ kg}$$

$$= 0.00055 \text{ u}$$

$$a_0 = 5.29 \times 10^{-11} \text{ m} \quad \text{Bohr radius (Hydrogen atom)}$$

$$hc = 1240 \text{ eV nm}$$

Solve for motion using Energy conservation

Potential energy and electric potential:

$$\Delta U_E = q\Delta V$$

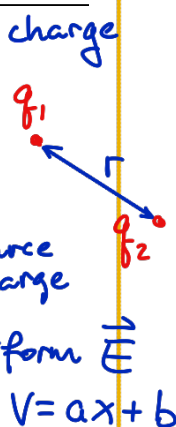
$$\Delta U_E = -qE_x\Delta x$$

$$U_E = \frac{kq_1q_2}{r}$$

$$V = \frac{kq}{r}$$

$$E_x = -\frac{\Delta V}{\Delta x}$$

$$\Delta K + \Delta U_E = W_{\text{ext}}$$



Resistance and circuits:

$$I = \frac{dq}{dt}$$

$$|\Delta V| = IR \quad \text{Ohm's law}$$

$$R = \frac{\rho L}{A}$$

$$P = I\Delta V$$

$$P_R = I^2 R = V^2/R$$

$$\Delta V_{\text{loop}} = 0$$

$$I_{\text{in}} = I_{\text{out}}$$

$$R_{\text{eq}} = R_1 + R_2 + R_3 + \dots$$

$$R_{\text{eq}} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots}$$

RC circuits:

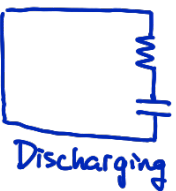
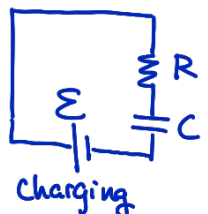
$$\tau = RC$$

$$Q(t) = Q_0 e^{-t/\tau} \quad \text{Discharging}$$

$$Q(t) = C\mathcal{E} (1 - e^{-t/\tau}) \quad \text{charging}$$

$$\Delta V_C = Q/C$$

$$I = I_0 e^{-t/\tau} \quad \text{Discharging}$$



Series
 $I_1 = I_2 = I_3$
 $V_1 + V_2 = V_{12}$
 parallel
 $V_1 = V_2 = V_{12}$
 $I_1 + I_2 = I_{12}$

Electric fields and forces:

$$\vec{F} = q\vec{E} \quad \text{test charge}$$

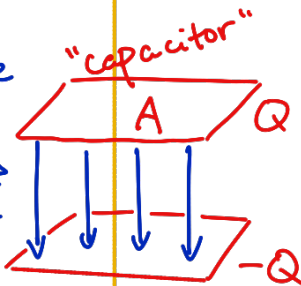
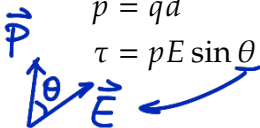
$$F = \frac{k|q_1||q_2|}{r^2}$$

$$E = \frac{k|q|}{r^2} \quad \text{source charge}$$

$$E = \frac{Q}{\epsilon_0 A}$$

$$p = qd$$

$$\tau = pE \sin \theta$$



Constant acceleration kinematics:

$$a_x = \frac{qE_x}{m} \quad \vec{F} = m\vec{a} = q\vec{E}$$

$$\Delta x(t) = v_{0x}t + \frac{1}{2}a_x t^2$$

$$v_x(t) = v_{0x} + a_x t$$

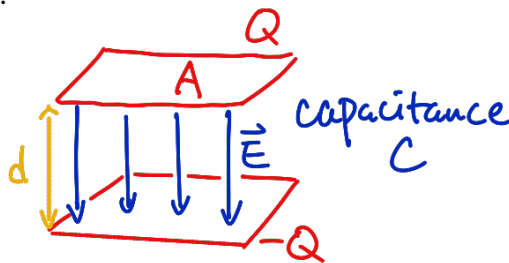
Capacitors and dielectrics:

$$Q = C\Delta V$$

$$\Delta V_C = Ed$$

$$C = \frac{\kappa\epsilon_0 A}{d}$$

$$U = \frac{1}{2}C(\Delta V)^2 = \frac{1}{2C}Q^2$$



$\kappa > 1$ if you insert dielectric material

Solve for motion of charged particles using kinematic equations.

angular frequency.

Magnetic forces and fields:

e⁻ motion

$$\vec{F}_B = q\vec{v} \times \vec{B}$$

$$F_B = |q|vB \sin \theta$$

$$R = mv / (|q|B)$$

$$f = \frac{|q|B}{2\pi m}$$

$$\vec{F}_B = L\vec{I} \times \vec{B}$$

$$F_B = ILB \sin \theta$$

$$F = \frac{\mu_0 I_1 I_2 L}{2\pi d}$$

$$B = \frac{\mu_0 I}{2\pi r}$$

$$B = \mu_0 NI / L$$

$$B = \frac{\mu_0 I}{2R}$$

$$m = IA$$

$$\tau = mB \sin \theta$$

Induction:

$$\mathcal{E} = vBL \quad \text{motional emf}$$

$$\Phi = BA_{\text{eff}} = BA \cos \theta$$

$$\mathcal{E} = N \left| \frac{d\Phi}{dt} \right| \quad \text{transformer emf}$$

Lenz's law: induced current opposes change in flux.

Right hand rules:

- RHR for cross product $\vec{C} = \vec{A} \times \vec{B}$: Point fingers with $\vec{A}$, curl fingers/face palm toward $\vec{B}$, thumb direction is $\vec{C}$

- RHR for loops: Curl fingers in direction of current, thumb points in direction of $\vec{B}$

EM waves:

$$\lambda f = c$$

$$E = cB$$

$$I = \frac{P}{A} = \frac{P}{4\pi r^2}$$

$$I = \frac{1}{2} \epsilon_0 c E_0^2$$

$$\vec{v} \parallel \vec{E} \times \vec{B}$$

$$A = 4\pi r^2$$

Quantum & atomic physics:

$$E = hf = \frac{hc}{\lambda}$$

$$\lambda = \frac{h}{mv}$$

$$E_n = \frac{h^2 n^2}{8mL^2}$$

$$E_n = -\frac{13.6 \text{ eV}}{n^2}$$

$$|L| = \sqrt{\ell(\ell+1)}\hbar$$

total projection on z-axis

Nuclear physics:

$$\Delta E = \Delta mc^2 \quad E = E_{\text{rest}} + KE$$

$$B(^A X) = [Zm_H + Nm_n - m(^A X)] c^2$$

$$t_{1/2} = \ln 2 / \lambda = \tau \ln 2 \quad \text{half-life}$$

$$N(t) = N_0 e^{-t/\tau} = N_0 2^{-t/t_{1/2}} \quad \tau = 1/\lambda$$

$$R(t) = \left| \frac{dN}{dt} \right| = N_0 e^{-t/\tau} \cdot \frac{1}{\tau} = \frac{\lambda N_0}{R_0} e^{-t/\tau}$$

$$R(t) = R_0 e^{-t/\tau} = R_0 2^{-t/t_{1/2}}$$

$$\text{initial amount } N_0 \rightarrow \text{final amount } N(t) + e^{-1/\alpha} \frac{1}{\tau}$$

$$E_i = KE_i + PE_i = E_f = KE_f + PE_f$$

$$\frac{1}{2} m v_i^2 + 0 = \frac{1}{2} m v_f^2 + e(0.25 \text{ V})$$

